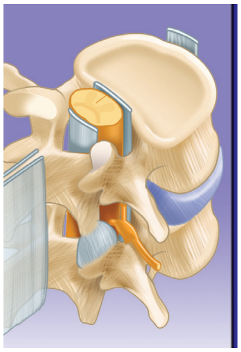
129

Traumatic Injuries of the Thoracolumbar Spine in the Athlete



Nourbakhsh Ali, Anuj Singla

Trauma to the thoracolumbar spine in athletes is common and can vary from minor sprain to major spinal fracture. Severe and potentially unstable spinal injuries are uncommon in sports-related trauma, but may be seen in athletes participating in contact sports or high velocity accidents. In addition to the contact and collision sports, athletes participating in certain noncontact sports like gymnastics, diving and weight lifting are also at risk for back trauma. Treatment of the injury and the recovery thereafter also varies widely depending on the pathology.

Irrespective of the underlying pathology, acute back pain is the most common symptom related to back trauma in athletes. The incidence of low back pain in athletes has been reported to be as high as 30%, and it can affect all athletes regardless of the position or the intensity of play.[1] Internal disc disruption is the most common cause of low back pain in young athletes. Lumbar disc disruption can cause a variety of symptoms, including back pain to radicular leg pain.[2–7] Some specific spine injuries are more common in certain sports (Table 129.1). Spondylolysis is more common in adolescent athletes, whereas internal disc disruption is more common between the ages of 18 and 50, and facet joints as well as sacroiliac (SI) joints more commonly caused back pain after the age of 50.[8] This chapter will cover the risk factors, history, examination, and imaging in the diagnosis of traumatic thoracolumbar spine injuries, and review the decision-making process and return-to-play (RTP) consideration in these athletes.

Risk Factors

During growth spurt the rate of bone growth surpasses the pace of muscle and ligaments growth, which leads to muscle imbalance and inflexibility.[9] Overuse injuries are more common during this period.[10] Avulsion of the ring apophyses as a secondary ossification center can happen due to repetitive flexion exercises. Disc can also herniate through the ring apophysis. Spina bifida or dysplasia of the posterior elements is a risk factor for spondylolysis.[11] Less skeletally mature children are at increased risk of spine injury when playing contact sports with physically larger players.[12]

Hyperlordosis of the lumbar spine due to hip flexor or hamstring tightness, genu recurvatum, increased femoral anteversion, abdominal muscle weakness, or increased thoracic kyphosis can also exert extra stress on the posterior spinal elements.[10] Other risk factors include malalignment of the lower extremities and pelvis, weak lower abdominal muscles, and abnormal pelvis and sacral morphology.[13,14] Increased pelvis incidence increases to a shear forces across the L5 pars interarticularis. Decreased pelvis incidence can also damage the L5 pars through the nutcracker mechanism. Both situations can result in spondylolysis and progress to spondylolisthesis. Sex-specific spine injury and pathologies are summarized in Table 129.2.

HISTORY

A complete history of the injury that led to the onset of back pain, the mechanism of injury, location and radiation of pain, exacerbating or alleviating activities, the type of sport(s) that the athlete has been involved in, the training regimen, any recent increase in the volume or intensity of training, the position played, diet, menstrual history, previous injuries, and treatments should be obtained.[10] Red flags include constitutional symptoms, history of infection, or cancer, weight loss, night sweats, fevers, chronic cough, immunocompromised patients, intravenous drug use, recent infection, and nonmechanical back pain worse at night. Urinary or stool incontinence should raise the concern about myelopathy or cauda equina syndrome. Cauda equina syndrome might also include lower extremity weakness/numbness, saddle anesthesia, and sexual dysfunction. Myelopathy can be caused by vertebrae fracture, epidural hematoma and retropulsion, or disc herniation at the level of the cervical or thoracic spine. The symptoms include ataxia, motor weakness, and in cases of cervical spinal cord impingement, the patient will have problems with the coordination of the hands.[15] Performance-enhancing steroids might compromise immune system and increase the propensity to infection. Wheelchair-bound athletes might also develop spine infections as a result of decubitus ulcer or recurrent urinary tract infection.[16]

PHYSICAL EXAMINATION

Physical exam should start with inspection of the back and extremities, palpation of the spine, and evaluation of the gait and range of motion (ROM) of the spine. Neurologic exam includes testing the strength, sensation, and reflexes. Nerve root impingement will cause radicular pain, possible sensory loss, weakness, or diminished reflexes.[17] Looking at the level of the shoulders and pelvis as well as the Adam forward bending test can help identify scoliosis. Skin abnormalities such as hemangiomas, café-au-lait spots, skin dimples, or hairy patches may

Abstract

Athletes are a high-risk group to spinal trauma because of increasing competitiveness, rigorous training, and decreased time to rest.

Spinal trauma can vary from nonspecific back pain to highly unstable spinal fractures. Details of the traumatic event and thorough evaluation of the involved athletes are paramount and can help the athlete return to his or her desired activities.

This chapter describes some of the common spinal trauma related pathologies and their treatment in detail, along with the general back trauma management guidelines.

Keywords

thoracolumbar trauma
athletes
spine fractures
spondylolysis
spondylolisthesis

TABLE 129.1 Specific Spine Injuries in Certain Sports

Sports	Spine Injury/Pathology
Rowers, and male football players, offensive and defensive linemen	Disc herniation
Baseball players and swimmers, gymnasts, weightlifters	Lumbar disc degeneration
Linemen, gymnasts, weightlifters, track and field athletes, and soccer players, dance, figure skating	Isthmic spondylolisthesis, spondylolysis
Gymnastics, wrestling, volleyball, and weight lifting	Vertebral body apophyseal avulsion fracture
Joggers	Stress fracture of the lamina
Ballet dancers	Fracture of the pedicle
Ballerina	Facet stress fractures
Classic ballet, rhythmic gymnastics, swimming, throwing, and serving, single arm throwing sports	Minor spine curves, scoliosis

TABLE 129.2 Spine Injuries in Males Versus Females

Male	Female
Scheuermann kyphosis	Spondylolysis in ballet, figure skating, and gymnastics
	Spondylolisthesis (higher in children of European descent than in those of African descent)
	Scoliosis
	Sacral stress fractures in long-distance runners

be associated with underlying spine abnormalities.[12] The positive straight leg raise (SLR) test is the provocation of radicular pain after 30 to 60 degrees of hip flexion while the patient is in the supine or sitting position.[18]

Pain with spine flexion can suggest injury to the anterior elements of the spine, including vertebrae fractures or muscle strain or disc disruption. Limitation of the spine forward flexion might be due to tight hamstrings. Pain with extension might indicate injury to the posterior spinal elements or SI joint.[11]

IMAGING

In the absence of red flag signs and after a thorough history and physical exam, the treatment of the athlete can be started without obtaining spine images, especially in cases of nonspecific acute low back pain.[19–22] It is very important to avoid any unnecessary radiation to young athletes.[23] Spine imaging is more appropriate when there is a history of trauma, and history of spine abnormalities or spondylolysis, or if the low back pain failed to respond to nonoperative management for 6 weeks or more.[24] Magnetic resonance imaging (MRI) is indicated in the presence of red flag signs, positive neurologic findings, history of infection or metastases, symptoms refractory to nonoperative management, and planning interventional procedures.[19,23]

DECISION-MAKING PRINCIPLES

Thoracolumbar Fractures

Thoracolumbar junction is the most common site for thoracic and lumbar spinal trauma. The most common primary force vectors responsible for spinal fractures include axial compression, lateral compression, flexion, extension, distraction, shear, and rotation. These individual vectors or their combinations can explain most of the commonly seen spinal injuries.

Spinal Stability

Spinal stability is one of the most important treatment determining factors in case of spinal trauma. It is described in terms of mechanical spinal stability and neurologic stability. Mechanical spinal stability defines the structural integrity of the spine and its ability to resist deforming forces. The Dennis classification system[24] for three column theory is commonly used for mechanical spinal stability. Anterior column is composed of the anterior longitudinal ligament (ALL), anterior vertebral body, and anterior annulus. The middle column comprises the posterior vertebral body, posterior annulus, and posterior longitudinal ligament (PLL), while the ligamentum flavum, neural arch, facet joints/capsule, and posterior ligamentous complex constitute posterior column. Involvement of two or more columns is considered mechanically unstable and may warrant stabilization. The presence or absence of neurologic deficit defines the neurologic stability of the spine.

The Thoracolumbar Injury Classification and Severity Score (TLICS)[25] is one of the commonly used classification systems that incorporates concepts of both mechanical and neurologic stability to help guide surgical versus conservative management of spinal fractures. Three injury characteristics, including (1) morphology of injury determined by radiographic appearance, (2) integrity of the posterior ligamentous complex, and (3) neurologic status of the patient, are analyzed and scored. A total score of 5 or more is considered an indication for surgical spinal stabilization (Table 129.3).

Compression Fractures

Compression fractures, though common in the general aging population, are relatively rare in young athletes. These fractures result from axial compression force with the resultant anterior column failure. It may involve superior/inferior or both end plates or anterior cortex buckling (Fig. 129.1). These fractures are usually mechanically stable and rarely neurologically involved. Plain radiographs/CT imaging is generally sufficient, and MRI is indicated only in suspected posterior complex/neurologic involvement. The prognosis is usually favorable and may be treated with observation/bracing. Surgical stabilization may be considered only in case of severe height loss (more than 50%) and significant kyphosis (more than 30 degrees), which is suggestive of potential posterior ligamentous complex involvement and possible spinal instability.[26]

Burst Fractures

Burst fractures are a type of compression fracture related to high-energy axial loading spinal trauma that results in disruption of the posterior vertebral body cortex with retropulsion into the spinal canal (Fig. 129.2). Distinction between compression and burst fractures may be difficult on plain radiographs and CT scan can help. MRI can provide substantial information on the involvement of the posterior ligamentous complex.

These fractures may be stable or unstable and can have varying degree of neurologic involvement.[27] Considerations for surgery include neurologic and mechanical instability. Neurologic involvement may require surgical decompression and stabilization. These fractures may be amenable to bracing/surgical stabilization, depending on the stability of spinal column.

TABLE 129.3 Thoracolumbar Injury Classification and Severity Score

Type	Qualifiers	Points
Injury Morphology		
Compression		1
Burst		2
Translational/rotational		3
Distraction		4
Integrity of Posterior Ligamentous Complex		
Intact		0
Indeterminate		2
Injured		3
Neurologic Status		
Intact		0
Nerve root		2
Cord/conus	Complete	2
	Incomplete	3
Cauda equine		3

Chance Fractures

Chance fractures are caused by flexion-distraction force vectors and involve axial compression of anterior column with transverse/distraction injury to middle/posterior columns (Fig. 129.3). Posterior column involvement could be bony or ligamentous. Ligament injuries may be evident by widening of facet joints/interspinous space on radiographs/CT. MRI is usually indicated and can identify the ligamentous involvement. These injuries are common with restrained car lap seatbelts (seatbelt fractures) and commonly associated with intra-abdominal injuries. Bony Chance fractures may be considered for bracing, while ligamentous involvement of posterior column may usually require surgical stabilization.

Fracture-Dislocations

These are highly unstable injuries caused by combination of various force vectors commonly seen in high energy trauma. All three spinal columns are involved with resultant spinal instability (Fig. 129.4). Facet joint disruption (fracture/dislocation) is associated with vertebral body translation. Spinal cord/conus with neurologic injuries is involved in 50% to 60% of these injuries. These injuries are typically both neurologically and mechanically unstable and require surgical intervention.

Extension/Extension-Distraction Injuries

These are relatively uncommon injuries and are usually seen in ankylosed spine associated with ankylosing spondylitis (AS) or diffuse idiopathic skeletal hyperostosis (DISH).

Stress Fractures

In addition to traumatic acute fractures, sudden unaccustomed increase in the training volume can cause stress fractures. Stress fractures of the spine represent 0.4% to 0.6% of all the stress fractures. Risk factors for stress fractures among athletes include low body mass index, endurance running, increase in the training intensity, and low bone mineral density.[16] Pedicle stress fracture or "pediculolysis" can happen in association with a contralateral spondylolysis.[28–30] Stress facet fractures have been reported with repetitive hyperextension activities, including gymnastics and

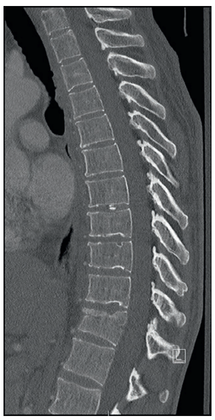


Fig. 129.1 Compression fracture.

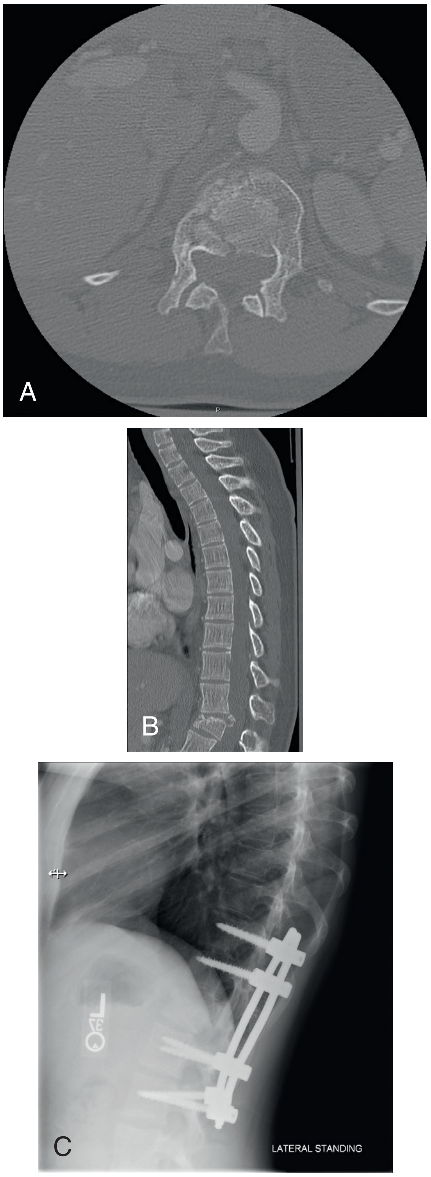


Fig. 129.2 Burst fracture. (A) and (B) Sagittal and axial computed tomography showing Burst fracture at T12. (C) Post spinal stabilization.

ballet dancing, and can usually be treated nonoperatively with rest and bracing.[16,31]

Vertebral Body Apophyseal Avulsion Fracture

Apophyseal ring fractures or limbus fractures are exclusively seen in skeletally immature adolescents. The apophyseal ring ossifies at the age of 4 to 6 and fuses to the body of the vertebrae at the age of 18. It is firmly attached to the annulus fibrosis, which is stronger than the osteocartilaginous junction. Following a compression or traction force, the apophyseal ring can be avulsed and retropulsed into the spinal canal. Apophyseal ring fractures happen at L4-L5 in more than 90% of cases.[32,33] Injuries to the apophyseal ring might happen during repetitive flexion and extension exercises. There is usually no neurologic deficit. The flexion and extension of the spine is limited; there is paraspinal muscle spasm. Computed tomography can better delineate the retropulsed bony fragments compared with MRI. Management usually starts with rest and anti-inflammatories. In case of significant neurologic deficit, the fragment needs to be excised surgically.[12]

Strains, Sprains, and Contusions

Strains, sprains, and contusions are the diagnosis of exclusion. Sprains happen due to the stretching of the ligaments beyond their elastic limit, and strains are muscle tears due to concentric or eccentric loading. Blunt impact to the soft tissues causes contusion and might result in hematoma formation and pain. This can cause spasm and localized tenderness. Hematoma or edema

Fig. 129.3 Chance fracture. (A) and (B) Sagittal and axial views showing 3 column flexion distraction fracture.

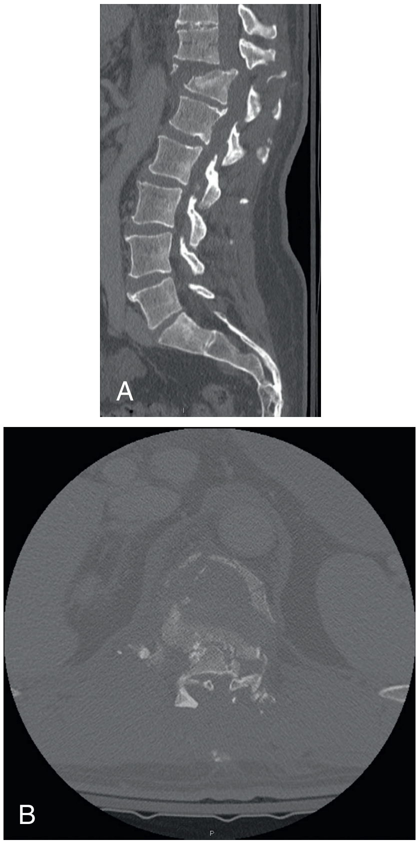


Fig. 129.4 Fracture dislocation. (A) and (B) Sagittal and axial images of fracture dislocation with severe spinal canal stenosis.

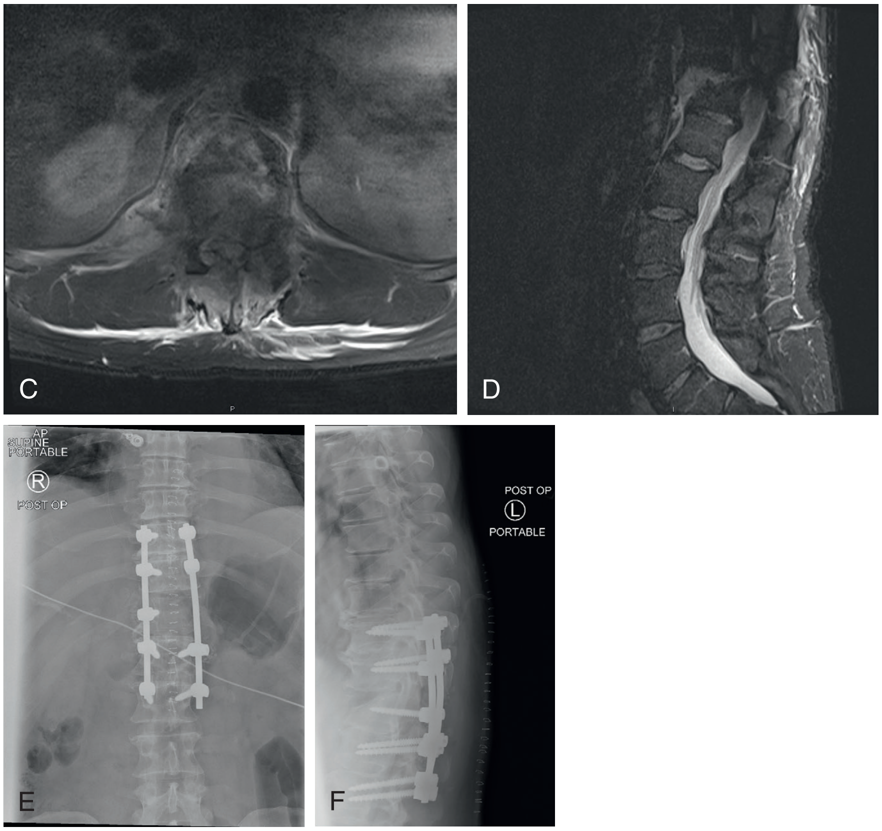


Fig. 129.4, cont'd (C) and (D) Magnetic resonance imaging images showing severe cord contusion. (E) and (F) Post spinal decompression and stabilization.

can be seen on the MRI scan. Management should start with nonsteroidal antiinflammatory drugs (NSAIDs), activity modification, and icing followed by progressive rehabilitation.[32]

Disc Herniation

Herniation of disc and subsequent impingement of the nerve roots in the lateral recess or the neural foramen can cause back pain with varying degree of radicular pain. Symptomatic disc herniations are located at L4-L5 and L5-S1 in 90% of cases.[34,35] Valsalva maneuver and coughing can increase pain. SLR can aid the diagnosis. As previously mentioned, MRI is not necessary in the acute phase of radicular symptoms with stable exam without neurologic deficit. If the symptoms are progressive or refractory, MRI needs to be obtained. The first step in treatment includes physical therapy and extension-based stabilization program, NSAIDs, and occasionally epidural steroid injections. Surgery is indicated in cases of progressive neurologic deficit, refractory pain, and cauda equina syndrome (Fig. 129.5). After regaining the full pain-free ROM of the lumbar spine, full strength in the lower extremities, and completion of a sports specific therapy, athletes may return to their previous activity.[12]

Spondylolysis

Spondylolysis (Fig. 129.6) is a defect in the pars interarticularis, which may be caused by repetitive torsion and extension of the spine and less commonly acute trauma. Spondylolysis has been seen in up to 47% of young athletes with back pain, more commonly at L5 on the left side.[8,36] Spondylolysis has equal prevalence among boys and girls; however, historically it was thought to be more common in boys.[14,37] It is more common in gymnasts and football linebackers due to repetitive hyperextension of the lumbar spine.[16]

Most common presenting symptoms include chronic refractory back pain, especially on repetitive activity and spinal extension. Neurologic symptoms including radiculopathy are not usual in the absence of associated spondylolisthesis. Incidental radiographic finding of pars defect in a completely asymptomatic athlete is also not uncommon.

Spondylolisthesis

Spondylolisthesis is the forward translation of rostral over the caudal vertebrae (Fig. 129.7). Spondylolisthesis can be classified as: dysplastic (I), isthmic (II), degenerative (III), traumatic (IV), and pathologic (V). The dysplastic type, though less common, has 32% chance of progression compared with 4% in the isthmic type.[38]

Symptoms include extension-related low back pain and pain with impact, such as during jumping. There is also decreased hamstring flexibility.[10] If spondylolisthesis causes neuroforaminal stenosis, the athlete may also complain of radicular symptoms and signs. Examination might reveal hyperlordosis of the lumbar spine, muscle spasm, hamstring tightness, and pain with extension of the spine and single-legged hyperextension test, which localizes the side of spondylolysis when the patient stands on the ipsilateral leg.[10]

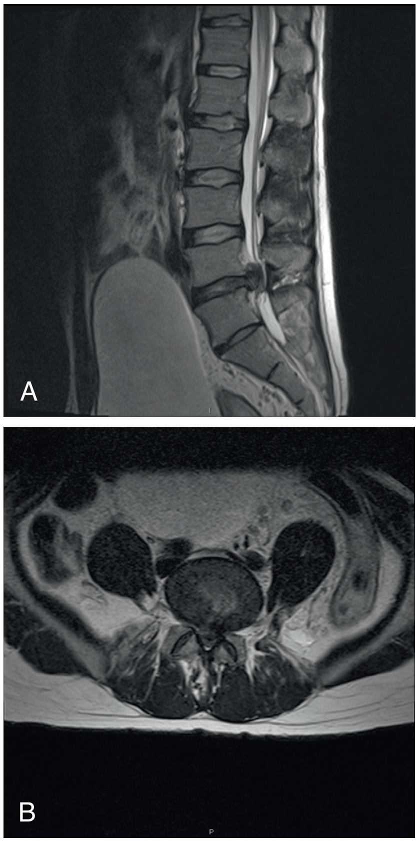


Fig. 129.5 Lumbar disc herniation. (A) and (B) L5-S1 disc herniation with severe central canal stenosis.

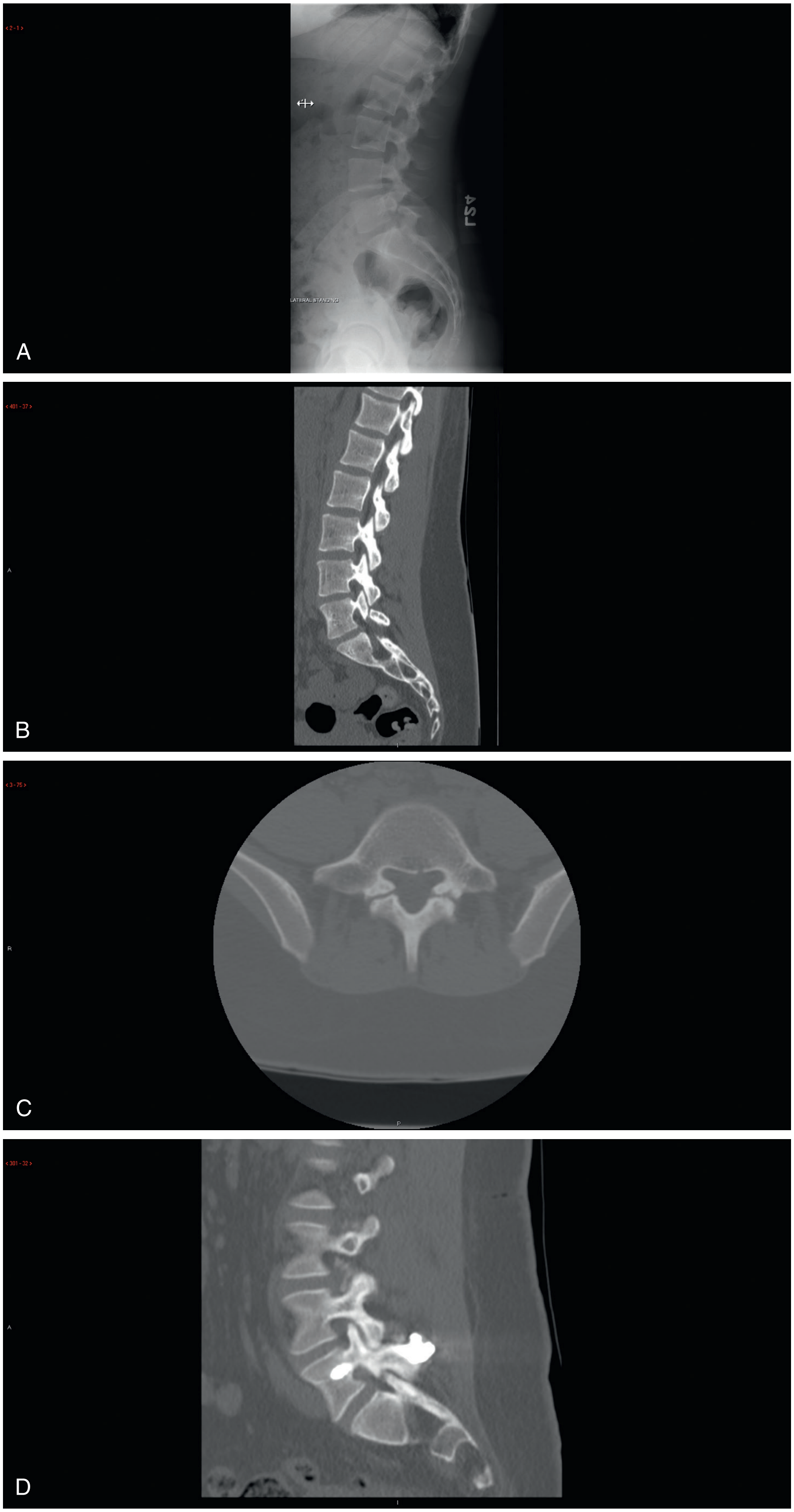


Fig. 129.6 Spondylolysis. (A) and (B) Lateral radiograph and Sagittal computed tomography (CT) images showing L5 pars defect. (C) Axial CT image. (D) Sagittal and (E) axial CT images showing healed pars defect.

Continued

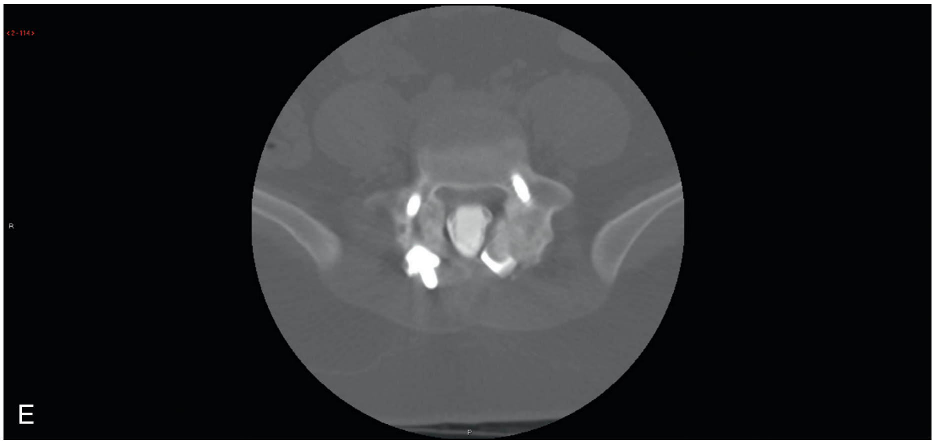


Fig. 129.6, cont'd

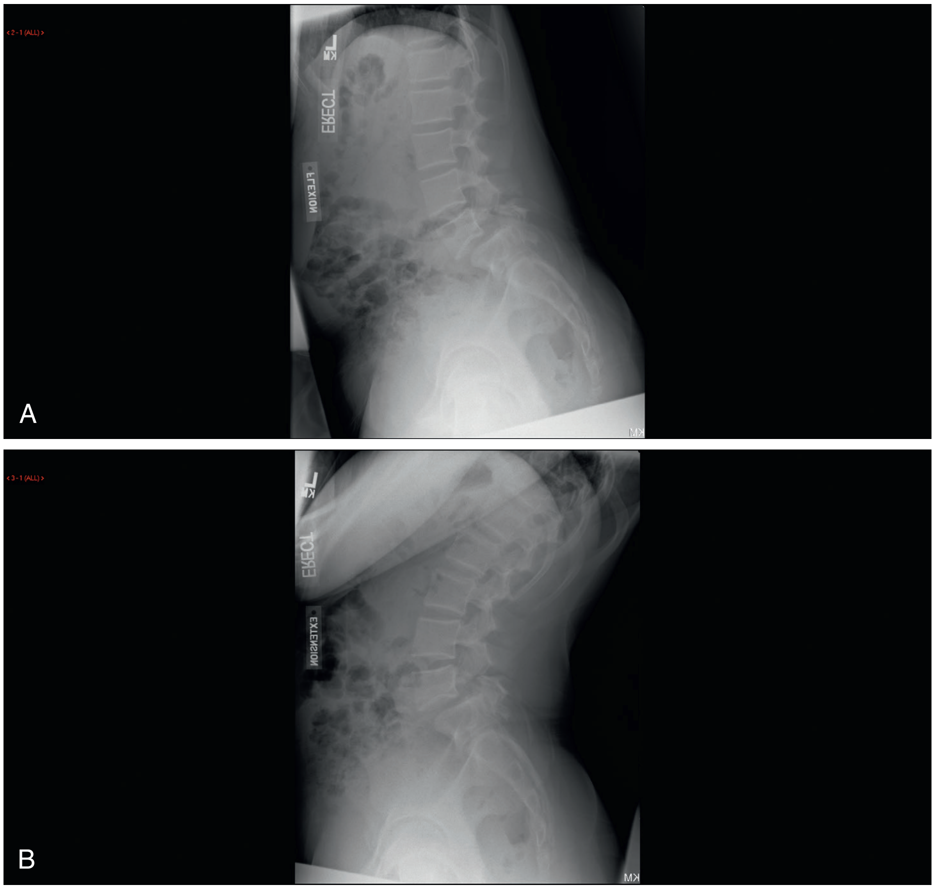


Fig. 129.7 Spondylolisthesis. (A) and (B) Flexion and extension stress images showing L5 pars defect and spondylolisthesis.

The AP view of the lumbar spine might show developmental defects such as spina bifida. The lateral view might show the lytic lesion in the pars interarticularis (see Fig. 129.6) or spondylolisthesis. The "neck of the Scottie dog" lesion is a stress reaction of the pars interarticularis on the oblique views. Because of the high amount of radiation dose, many institutions discourage the routine use of oblique views. Only one-third of the stress fractures can be seen on plain x-rays.[2] On the single-photon emission computed tomography (SPECT) bone scan, active bony lesions over shows up as an area with increased uptake. Computed tomography confirms the presence of pars interarticularis stress fracture and also can be used to monitor the healing process.[9]

Posterior Element Overuse Syndrome

Repeated extension and rotation of the spine can cause posterior element overuse syndrome, which involves musculotendinous junction, ligaments, and facet joints. It is also called *hyperlordotic low back pain*. This is the most common cause of low back pain in adolescents after spondylolysis.[2] The symptoms include pain with extension of the spine, paraspinal muscle tenderness.

Management consists of antiinflammatories and icing. Rehabilitation should start with pain-free activities, and preventing hyperextension, abdominal strengthening, antilordotic exercises, thoracolumbar, and hamstring stretches should be included in the rehabilitation regimen.[10]

Sacroiliac Joint

The SI joint pain can be a cause of low back pain in athletes, and it might radiate to the lower extremity with variable patterns. It is more commonly seen in cross-country skiers and rowers and in athletes who perform single leg stance activities like gymnastics, skating, and bowling. It can result from axial loading and rotation, direct trauma shearing forces across the joint, enthesopathy, and ligamentous injuries, scoliosis, and leg length discrepancy. SI joint pain can be due to infection, inflammation such as seronegative spondyloarthropathy, or Crohn disease. Stress fracture of the sacrum can also cause SI joint pain.[10] There are no definitive exacerbating or relieving movements that can be specifically attributed to SI joint pain. Patients usually complain of a midline back pain.[2] On exam, pain is localized to the buttock or lumbar region and worsens with spine extension. If inflammation or infection is suspected, erythrocyte sedimentation rate, C-reactive protein, rheumatoid factor, antinuclear antibody, and HLA-B27 can be obtained.[10]

Treatment Options

The myriad spinal conditions, injuries, and surgical options highlight the need to evaluate spine injuries according to each specific injury and its respective treatment modality. Activity should be limited till complete evaluation of any serious underlying injury. In the absence of red flags, unstable spine, or major trauma, most athletes respond favorably to conservative treatment. The natural course of acute low back pain is usually favorable, and the athlete should be encouraged to remain active. This will increase the likelihood of quicker recovery when compared with bedrest.[24] Most episodes of low back pain resolve in 2 to 4 weeks.[39] Biomechanical studies have shown that effective control of multifidus and transversus abdominus can increase the segmental stability of the spine. Core muscle strengthening can prevent the recurrence of low back pain.[40] This has been shown to be more effective than education and medical management for the treatment of low back pain.[39,41] Short-term outcomes have been reported from epidural steroid injections for discogenic back pain.[42,43]

The use of lumbar corsets, braces, and traction as treatment options for low back pain is not supported in the literature.[3,44] If the patient fails to respond to nonoperative management for 6 weeks, further imaging and possibly more invasive procedures like epidural injections are recommended. If surgery is delayed, it does not result in worse neurologic outcomes.[45] The recurrence rates of low back pain has been reported to be between 50% and 84% in the general population.[46–48]

Sacroiliac Joint

Treatment starts with rest, antiinflammatories, and physical therapy, as well as manipulation. Rehabilitation should target core strengthening and the lumbosacral, pelvis, and gluteal regions.[2] In case of sacral stress fracture, protected weight bearing is recommended until the pain resolves.[10] If the improvement reaches a plateau, SI joint injections can be therapeutic as well as diagnostic.[2]

Spondylolysis and Spondylolisthesis

The first line of treatment includes activity modification, physical therapy, or bracing for 3 to 6 months. The goal of physical therapy is to decrease lumbar lordosis, along with core muscle strengthening and hamstring stretching. Activities that cause pain should be avoided—especially extension activities. Strengthening of the hip flexors, hamstring, and abdominal muscles are recommended. A lumbar brace can decrease the amount of lordosis and decrease the pressure on the pars interarticularis. An example of this is the Boston brace, which is molded in 0 to 15 degrees of flexion.

RTP after nonoperative management is allowed when the patient achieves pain-free ROM of the lumbar spine. The activity needs to be gradually increased as the patient goes back to sports. If bracing was used, it should be continued until the athlete resumes full painless activity. At that point, the brace can be weaned off gradually. Nonoperative management has been shown to be successful in providing good to excellent clinical results in 78% of patients, and up to 89% of patients return to sports.[16,49–53] Unilateral lesions and half of bilateral lesions are reported to heal better; however, healing rarely occurs in chronic lesions and dysplastic spondylolisthesis. In most patients, fibrous union can provide acceptable stability.[32,54,55]

When the patient resumes full painless activity out of the brace, the spondylolysis/spondylolisthesis is considered clinically healed. The athlete should be followed every 4 to 6 months with standing lateral films on until skeletal maturity to monitor the amount of slip progression. Surgical stabilization is considered when there is more than 50% of slip, in the presence of neurologic symptoms, or in cases of persistent pain.[9] Surgical procedures include pedicle screw and hook technique, the Buck technique with translaminar screws, and spinous process or transverse process wiring.

Fractures

The guidelines for treatment of stable or unstable spine fractures in athletes are the same as nonathletes. In general, a treatment plan needs to be individualized for any spinal fracture. Immediate goals of treatment include stabilizing the spine (observation/bracing vs. surgical stabilization) while the fracture heals. Surgical indications for spine fractures include neurologic compromise due to herniated disc, hematoma or fracture fragments, or mechanical instability, including fracture dislocations.[32] For most stable fractures, after the resolution of pain and evidence of radiologic union and a period of rehabilitation, athletes can return to sports activities.[56] For incomplete cord injuries, surgical intervention in the first 24 to 72 hours provides better neurologic outcomes compared with delayed surgery.[57,58]

Compression fractures, spinous process fractures, and transverse process fractures can usually be treated with bracing/observation, and the athletes can return to full activity after complete resolution of symptoms.[59]

POSTOPERATIVE MANAGEMENT AND RETURN TO PLAY

Despite a lack of consensus and specific recommendations, there is universal agreement that athletes should be pain free, completely neurologically intact, and have full strength and ROM before returning to play after spinal injury.[59,60] Gradual resumption of activity and extended conditioning is universally recommended. Allowing the athletes to return to contact sports is an even more difficult decision.

Athletes with stable fractures can usually return to full activity after complete healing. Spinal fusions that bypass transition zones in thoracolumbar region are an absolute contraindication to participation in contact sports.[60,61] Similarly, fusions that terminate at these transition zones represent a contraindication for RTP. However, players may RTP if a fusion does not cross transitional levels and they meet general criteria.[60,61] Successful conservative or surgical treatment of lumbar disk herniation is generally well accepted for return to full contact sports. A very high successful RTP rate (up to 81%)[62] has been reported for athletes with disc herniation. RTP after 2 to 6 months is plausible for contact sports after percutaneous discectomy and microdiscectomy and 4 to 8 weeks for lighter activities such as golf.[60]

RTP after surgical treatment of spondylolysis is controversial, and formal criteria are lacking. Guidelines do not recommend return to contact sports after fusion of spondylolysis and spondylolisthesis.[60]

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Huang P, Anissipour A, McGee W, et al. Return-to-play recommendations after cervical, thoracic, and lumbar spine injuries: a comprehensive review. *Sports Health*. 2016;8(1):19–25.

Level of Evidence:
IV

Summary:
Despite a lack of consensus and specific recommendations, there is universal agreement that athletes should be pain free, completely neurologically intact, and have full strength and range of motion before returning to play after spinal injury.

Citation:
Haus BM, Micheli LJ. Back pain in the pediatric and adolescent athlete. *Clin Sports Med*. 2012;31(3):423–440. doi:10.1016/j.csm.2012.03.011.

Level of Evidence:
IV

Summary:
Acute trauma or chronic overuse can contribute to the development of pain in athletes. The differential diagnosis of acute pain includes thoracolumbar fractures, acute disc herniation, muscular sprains, and apophyseal ring injuries. The overuse injuries include spondylolisthesis and spondylolysis, hyperlordotic mechanical back, and degenerative disc disease.

Citation:
Mautner KR, Huggins MJ. The young adult spine in sports. *Clin Sports Med*. 2012;31(3):453–472. doi:10.1016/j.csm.2012.03.007.

Level of Evidence:
IV

Summary:
Internal disc disruption is the most common cause of low back pain in young athletes. The natural course of acute low back pain is favorable in athletes. The focus of rehabilitation should be on core stability and strengthening of the hip girdle. Epidural steroid injections and more invasive procedures can be considered if the athlete does not respond to nonoperative management.

Citation:
Lawrence JP, Greene HS, Grauer JN. Back pain in athletes. *J Am Acad Orthop Surg*. 2006;14(13):726–735.

Level of Evidence:
IV

Summary:
Self-limiting symptoms, as opposed to persistent or recurrent symptoms, should be differentiated in athletes. Different kinds of sports place different loads on the spine and result in specific spinal pathologies. Physical conditioning can help avoid spine injuries in recreational athletes.

REFERENCES

1. Dreisinger TE, Nelson B. Management of back pain in athletes. *Sports Med*. 1996;21:313–320.
2. Mautner KR, Huggins MJ. The young adult spine in sports. *Clin Sports Med*. 2012;31:453–472.
3. Rhee JM, Schaufele M, Abdu WA. Radiculopathy and the herniated lumbar disc. Controversies regarding pathophysiology and management. *J Bone Joint Surg Am*. 2006;88:2070–2080.
4. Boos N, Weissbach S, Rohrbach H, et al. Classification of age-related changes in lumbar intervertebral discs: 2002 Volvo Award in Basic Science. *Spine*. 2002;27:2631–2644.
5. Coppes MH, Marani E, Thomeer RT, et al. Innervation of "painful" lumbar discs. *Spine*. 1997;22:2342–2349, discussion 2349–2350.
6. Brown MF, Hukkanen MV, McCarthy ID, et al. Sensory and sympathetic innervation of the vertebral endplate in patients with degenerative disc disease. *J Bone Joint Surg Br*. 1997;79:147–153.
7. DePalma MJ, Lee JE, Peterson L, et al. Are outer annular fissures stimulated during diskography the source of diskogenic low-back pain? An analysis of analgesic diskography data. *Pain Med*. 2009;10:488–494.
8. Micheli LJ, Wood R. Back pain in young athletes. Significant differences from adults in causes and patterns. *Arch Pediatr Adolesc Med*. 1995;149:15–18.
9. d'Hemecourt PA, Gerbino PG 2nd, Micheli LJ. Back injuries in the young athlete. *Clin Sports Med*. 2000;19:663–679.
10. Zetaruk M. Lumbar spine injuries. In: Micheli L, Purcell L, eds. *The Adolescent Athlete*. New York: Springer; 2007:109–140.
11. Purcell L, Micheli L. Low back pain in young athletes. *Sports Health*. 2009;1:212–222.
12. Simon LM, Jih W, Buller JC, et al. Back pain and injuries. In: Griesemer B, ed. *Pediatric Sports Medicine for Primary Care*. Philadelphia: Lippincott Williams & Wilkins; 2002: 306–325.
13. Kujala UM, Taimela S, Oksanen A, et al. Lumbar mobility and low back pain during adolescence. A longitudinal three-year follow-up study in athletes and controls. *Am J Sports Med*. 1997;25:363–368.
14. Marty C, Boisaubert B, Descamps H, et al. The sagittal anatomy of the sacrum among young adults, infants, and spondylolisthesis patients. *Eur Spine J*. 2002;11:119–125.
15. Quinones-Hinojosa A, Jun P, Jacobs R, et al. General principles in the medical and surgical management of spinal infections: a multidisciplinary approach. *Neurosurg Focus*. 2004;17:E1.
16. Metz LN, Wustrack R, Lovell AF, et al. Infectious, inflammatory, and metabolic diseases affecting the athlete's spine. *Clin Sports Med*. 2012;31:535–567.
17. Barr K, Harrast M. Low back pain. In: Braddom R, ed. *Physical Medicine and Rehabilitation*. Philadelphia: Saunders/Elsevier; 2007:883–927.
18. Deville WL, van der Windt DA, Dzaferagic A, et al. The test of lasegue: systematic review of the accuracy in diagnosing herniated discs. *Spine*. 2000;25:1140–1147.
19. Chou R, Qaseem A, Snow V, et al. Diagnosis and treatment of low back pain: a joint clinical practice guideline from the American College of Physicians and the American Pain Society. *Ann Intern Med*. 2007;147:478–491.
20. Espeland A, Baerheim A, Albrektsen G, et al. Patients' views on importance and usefulness of plain radiography for low back pain. *Spine*. 2001;26:1356–1363.
21. Atlas SJ, Deyo RA. Evaluating and managing acute low back pain in the primary care setting. *J Gen Intern Med*. 2001;16:120–131.
22. Kendrick D, Fielding K, Bentley E, et al. Radiography of the lumbar spine in primary care patients with low back pain: randomised controlled trial. *BMJ*. 2001;322:400–405.
23. Jarvik JG. Imaging of adults with low back pain in the primary care setting. *Neuroimaging Clin N Am*. 2003;13:293–305.
24. Denis F. The three column spine and its significance in the classification of acute thoracolumbar spinal injuries. *Spine*. 1983;8:817–831.
25. Vaccaro AR, Lehman RA Jr, Hurlbert RJ, et al. A new classification of thoracolumbar injuries: the importance of injury morphology, the integrity of the posterior ligamentous complex, and neurologic status. *Spine*. 2005;30:2325–2333.
26. Day B, Kokan P. Compression fractures of the thoracic and lumbar spine from compensable injuries. *Clin Orthop Relat Res*. 1977;124:173–176.
27. Junkins EP Jr, Stotts A, Santiago R, et al. The clinical presentation of pediatric thoracolumbar fractures: a prospective study. *J Trauma*. 2008;65:1066–1071.
28. Sairyo K, Katoh S, Sasa T, et al. Athletes with unilateral spondylolysis are at risk of stress fracture at the contralateral pedicle and pars interarticularis: a clinical and biomechanical study. *Am J Sports Med*. 2005;33:583–590.
29. Weatherley CR, Mehdian H, Berghe LV. Low back pain with fracture of the pedicle and contralateral spondylolysis. A technique of surgical management. *J Bone Joint Surg Br*. 1991;73:990–993.
30. Gunzburg R, Fraser RD. Stress fracture of the lumbar pedicle. Case reports of "pediculolysis" and review of the literature. *Spine*. 1991;16:185–189.
31. Fehlandt AF Jr, Micheli LJ. Lumbar facet stress fracture in a ballet dancer. *Spine*. 1993;18:2537–2539.
32. Haus BM, Micheli LJ. Back pain in the pediatric and adolescent athlete. *Clin Sports Med*. 2012;31:423–440.
33. Fakouri B, Nnadi C, Boszczyk B, et al. When is the appropriate time for surgical intervention of the herniated lumbar disc in the adolescent? *J Clin Neurosci*. 2009;16:1153–1156.
34. Gaunt AM. Caring for patients who have acute and subacute low back pain. *CME Bul*. 2008;7:1–8.
35. Lawrence JP, Greene HS, Grauer JN. Back pain in athletes. *J Am Acad Orthop Surg*. 2006;14:726–735.
36. Gregory PL, Batt ME, Kerslake RW, et al. Single photon emission computerized tomography and reverse gantry computerized tomography findings in patients with back pain investigated for spondylolysis. *Clin J Sport Med*. 2005;15:79–86.
37. Roche MB, Rowe GG. The incidence of separate neural arch and coincident bone variations; a summary. *J Bone Joint Surg Am*. 1952;34-A:491–494.
38. McPhee IB, O'Brien JP, McCall IW, et al. Progression of lumbosacral spondylolisthesis. *Australas Radiol*. 1981;25:91–95.
39. Hides JA, Richardson CA, Jull GA. Multifidus muscle recovery is not automatic after resolution of acute, first-episode low back pain. *Spine*. 1996;21:2763–2769.
40. Goel VK, Kong W, Han JS, et al. A combined finite element and optimization investigation of lumbar spine mechanics with and without muscles. *Spine*. 1993;18:1531–1541.
41. Ferreira PH, Ferreira ML, Maher CG, et al. Specific stabilisation exercise for spinal and pelvic pain: a systematic review. *Aust J Physiother*. 2006;52:79–88.
42. Manchikanti L, Cash KA, McManus CD, et al. Preliminary results of a randomized, double-blind, controlled trial of

fluoroscopic lumbar interlaminar epidural injections in managing chronic lumbar discogenic pain without disc herniation or radiculitis. *Pain Physician*. 2010;13:E279–E292.
43. Buttermann GR. The effect of spinal steroid injections for degenerative disc disease. *Spine J*. 2004;4:495–505.
44. Gay RE, Brault JS. Evidence-informed management of chronic low back pain with traction therapy. *Spine J*. 2008;8:234–242.
45. Buttermann GR. Treatment of lumbar disc herniation: epidural steroid injection compared with discectomy. A prospective, randomized study. *J Bone Joint Surg Am*. 2004;86-A:670–679.
46. Von Korff M, Deyo RA, Cherkin D, et al. Back pain in primary care. Outcomes at 1 year. *Spine*. 1993;18:855–862.
47. Troup JD, Martin JW, Lloyd DC. Back pain in industry. A prospective survey. *Spine*. 1981;6:61–69.
48. Pengel LH, Herbert RD, Maher CG, et al. Acute low back pain: systematic review of its prognosis. *BMJ*. 2003;327:323.
49. Sys J, Michielsen J, Bracke P, et al. Nonoperative treatment of active spondylolysis in elite athletes with normal x-ray findings: literature review and results of conservative treatment. *Eur Spine J*. 2001;10:498–504.
50. McCleary MD, Congeni JA. Current concepts in the diagnosis and treatment of spondylolysis in young athletes. *Curr Sports Med Rep*. 2007;6:62–66.
51. Steiner ME, Micheli LJ. Treatment of symptomatic spondylolysis and spondylolisthesis with the modified Boston brace. *Spine*. 1985;10:937–943.
52. George SZ, Delitto A. Management of the athlete with low back pain. *Clin Sports Med*. 2002;21:105–120.
53. McTimoney CA, Micheli LJ. Current evaluation and management of spondylolysis and spondylolisthesis. *Curr Sports Med Rep*. 2003;2:41–46.
54. Hollenberg GM, Beattie PF, Meyers SP, et al. Stress reactions of the lumbar pars interarticularis: the development of a new MRI classification system. *Spine*. 2002;27:181–186.
55. Newman PH. A clinical syndrome associated with severe lumbo-sacral subluxation. *J Bone Joint Surg Br*. 1965;47:472–481.
56. Baranto A, Hellstrom M, Cederlund CG, et al. Back pain and MRI changes in the thoraco-lumbar spine of top athletes in four different sports: a 15-year follow-up study. *Knee Surg Sports Traumatol Arthrosc*. 2009;17:1125–1134.
57. Vaccaro AR, Daugherty RJ, Sheehan TP, et al. Neurologic outcome of early versus late surgery for cervical spinal cord injury. *Spine*. 1997;22:2609–2613.
58. La Rosa G, Conti A, Cardali S, et al. Does early decompression improve neurological outcome of spinal cord injured patients? Appraisal of the literature using a meta-analytical approach. *Spinal Cord*. 2004;42:503–512.
59. Eddy D, Congeni J, Loud K. A review of spine injuries and return to play. *Clin J Sport Med*. 2005;15(6):453–458.
60. Huang P, Anissipour A, McGee W, et al. Return-to-play recommendations after cervical, thoracic, and lumbar spine injuries: a comprehensive review. *Sports Health*. 2016;8(1):19–25. [Epub 2015 Oct 14].
61. Burnett MG, Sonntag VK. Return to contact sports after spinal surgery. *Neurosurg Focus*. 2006;21(4):E5.
62. Hsu WK, McCarthy KJ, Savage JW. The professional athlete spine initiative: outcomes after lumbar disc herniation in 342 elite professional athletes. *Spine J*. 2011;11(3):180–186.